

Neural Network-Based Wildfire Risk Classification

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1 Introduction

Wildfires are a growing threat to infrastructure, ecosystems, and human safety. In addition to the immediate danger of fire, wildfire smoke can travel thousands of kilometers, degrading air quality and affecting millions of people [NASA \(2025b\)](#). Predicting the likelihood of wildfires in advance can help authorities, first responders, and communities with preparation, resource allocation, and reduce potential damage.

This project aims to classify regions into low, medium, and high wildfire risk using a machine learning model trained on environmental and meteorological data. The resulting risk predictions can support proactive decision making and reduce the overall impact of wildfires on both people and the environment.

1.1 Related Work

Predicting wildfires is a sought after problem in both environmental science and machine learning. Existing approaches can be categorized into deterministic models, statistical models, and machine learning models.

Deterministic models, such as the Canadian Forest Fire Danger Rating System (CFFDRS) rely on empirical formulae derived from field experiments to estimate wildfire risk based on meteorological factors [Government of Canada \(2025\)](#). These models are generally used in operational settings but are limited in their ability to capture non-linear interactions between variables.

Statistical models include regression-based and Bayesian approaches that predict wildfire likelihood using historical fire and weather data [Cisneros et al. \(2024\)](#); [Joseph et al. \(2019\)](#). These methods are effective in identifying trends but they are limited in their ability to capture rare or extreme weather conditions, which is critical for high-risk classification.

Machine learning approaches have used tree-ensemble methods such as XGBoost to classify wildfire risk based on tabular environmental data [Hu et al. \(2025\)](#). More recently however, neural networks have been trained on both satellite imagery and meteorological data to predict wildfire spread [Shadrin et al. \(2024\)](#). These models can learn complex, non-linear relationships but require preprocessing and normalization of inputs.

2 Dataset

For this project, we will utilize a Wildfire Prediction Dataset found on Kaggle [Vijayaragul \(2025\)](#). This dataset contains 118858 entries and 17 attributes, with no missing values. Further, all values are numerical (float64), simplifying any data manipulation by avoiding issues revolving around categorical data.

2.1 Data Sources

There are two main sources used by this dataset: NASA's FIRMS VIIRS SCC system for fire radiative power (FRP) data, and Open-Meteo's meteorological data [NASA \(2025a\)](#); [Open-Meteo \(2025\)](#). As these sources are open and publicly available, the data used is compliant with all terms of service.

2.2 Features

The dataset contains the following features:

- **daynight_N**: Indicator for whether the observation was taken during the day or night.
- **lat, lon**: Latitude and longitude coordinates of the location where the observation was recorded.
- **fire_weather_index**: An index representing the overall fire risk based on meteorological conditions.
- **pressure_mean**: Mean atmospheric pressure at the location.

- **wind_direction_mean, wind_direction_std:** Mean and standard deviation of wind direction.
- **solar_radiation_mean:** Average solar radiation received.
- **dewpoint_mean:** Average dew point temperature.
- **cloud_cover_mean:** Average cloud cover percentage.
- **evapotranspiration_total:** Total evapotranspiration over the observation period.
- **humidity_min:** Minimum humidity recorded.
- **temp_mean, temp_range:** Mean temperature and temperature range.
- **wind_speed_max:** Maximum wind speed observed.
- **occured:** Binary flag indicating whether a fire occurred (1) or not (0).
- **frp:** Fire Radiative Power, measuring the intensity of a fire if it occurred.

2.3 Preprocessing

The dataset requires minimal preprocessing due to its numerical format. The preprocessing steps used are:

- **Clipping negative values:** Negative *fire_weather_index* values are set to 0, as negative FWI is meaningless (i.e., negative risk is nonsensical).
- **Feature normalization:** All numeric features are standardized to facilitate stable training of the neural network. Each feature was scaled to have a mean of 0 and a standard deviation of 1, ensuring that features with larger ranges do not dominate the training process.

2.4 Data Split

A standard training/validation/testing split will be used, 70% for training, 15% for validation, 15% for testing.

3 Features

All 17 numerical features are used as input to the neural network. These features, described in Section 2.2, include positional, meteorological and environmental data. Positional features (*lat*, *lon*) provide the geographical context, important to the input as wildfire risk is often region-specific due to climate and vegetation. Meteorological features, such as *temp_mean*, *temp_range*, *humidity_min*, *pressure_mean*, *cloud_cover_mean*, *dewpoint_mean*, *solar_radiation_mean*, etc, describe the atmospheric conditions that directly influence fire ignition and propagation. Environmental features, including *evapotranspiration_total*, *fire_weather_index*, etc, describe the conditions that influence fire behaviour and ignition potential.

The model's target variable is the wildfire risk category, derived from the *fire_weather_index* and *occurred* variables in the dataset, classified into low, medium, or high risk. By combining multiple features that capture positional, meteorological, and environmental conditions, the neural network can learn complex, non-linear relationships that contribute to wildfire risk.

4 Implementation

The neural network, implemented in `02_model_training.ipynb`, uses a feed-forward MLP optimized for meteorological data. The model takes 16 input features and predicts the probability of belonging to one of the three wildfire risk classes. A combination of linear, ReLU, softmax and dropout functions were used in this model. Model training ran for up to 95 epochs with early stopping (patience = 5).

The final model implemented is a Neural Network consisting of 4 fully connected layers. The model achieved an accuracy of 0.9448 on our test set. The architecture is as follows:

- Input Layer: 16 Features
- Hidden Layer 1: 256 neurons
- Hidden Layer 2: 128 neurons
- Hidden Layer 3: 64 neurons
- Output Layer: 3 neurons

The activation function used is ReLU.

Regularization consisted of the following methods:

- L2 Regularization using weight decay of strength 1×10^{-4}
- Batch Normalization after each hidden layer
- Dropout on each hidden layer with a dropout rate of 0.3 for the first two layers and 0.2 for the last hidden layer

Comparing our model with a simple baseline of a majority vote strategy we achieved much higher accuracy. Since there are 3 classes and low is the most common the majority vote strategy entails voting for the ‘Low’ class each time. This results in a test accuracy of 0.4998. Our model significantly outperformed this benchmark.

The model utilized a cross entropy loss function. Since the number of samples in each class was uneven we applied a weighted cross entropy loss to ensure that minority classes have a higher weight during training to prevent the model from being biased towards the majority class. This is shown in Figure 1.

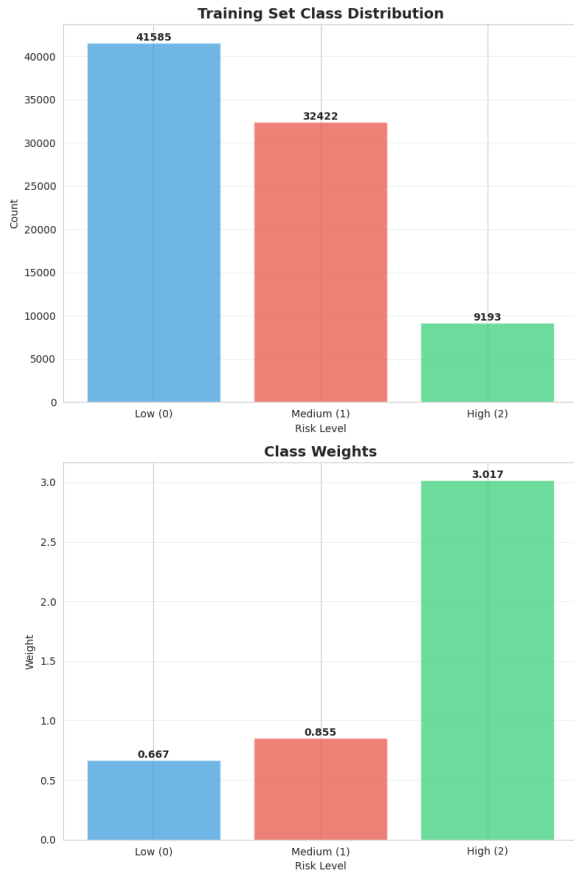
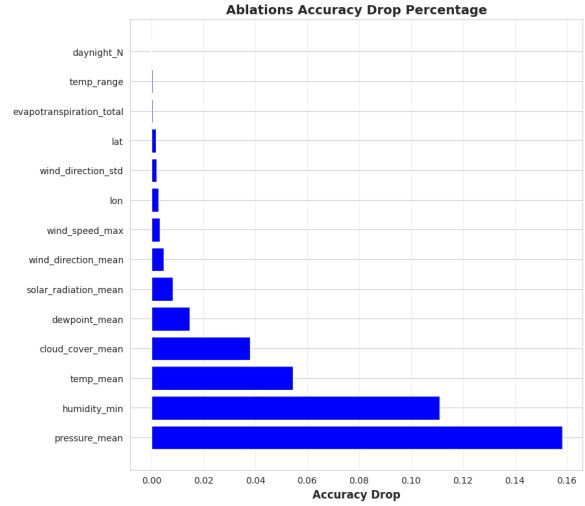


Figure 1: Weighted cross entropy loss applied to compensate for class imbalance.

The optimization function used is the AdamW optimizer. We used this optimization function over a traditional SGD optimizer since it experienced

faster convergence due to its parameter updating methodology which takes into account the moving averages of the mean and variance of the gradient as opposed to the SGD optimizer which updates based on the gradient of the loss.

The results of holding out specific features in the model is shown in Figure 2. This chart shows the effect on the accuracy of the model when holding out each specific feature. This shows that the most important features are pressure_mean and humidity_min.



Model accuracy when holding out individual features during training.

5 Results and Evaluation

Our model performs a three-way data split into training, test, and validation datasets. We used a stratified split to maintain the distribution of our target variable *risk_level* between the sets. The purpose of the splits are as follows:

1. **Training set:** used to fit the model parameters and to update the weights during backpropagation.
2. **Validation set:** used to determine performance during training and used for our early stop gradient descent and dynamic learning rate scheduling.
3. **Test set:** used to evaluate the model performance on unseen data and provide an accurate measurement of our model’s ability to generalize and predict on new data.

Our evaluation metrics are based on the classification performance. We computed the precision, recall, and F1-score for each risk class, together with the overall accuracy and a confusion matrix.

Accuracy serves as the primary evaluation metric, while the F1-score provides a more balanced measure of performance across all risk categories.

The neural-network classifier was trained on GPU for 95 epochs with early stopping. The model achieved a validation accuracy of 0.9439 before convergence and a test accuracy of 0.9432. The F1 scores show relatively good performance:

- **Low-risk (class 0):** F1 = 0.96
- **Medium-risk (class 1):** F1 = 0.93
- **High-risk (class 2):** F1 = 0.92

The confusion matrix shows that most misclassifications are between medium and high-risk wildfires. Precision and recall vary for the each class in the range of 0.94 to 1.00. Our metrics from the previous progress report were adequate, and with our modifications to the neural network, we achieved much greater accuracy and F1 scores.

6 Progress

In our progress report, we highlighted several key goals for the final iteration of the project. These included finalizing the data preprocessing module by properly cleaning unrealistic values in the `fire_weather_index`, analyzing the relative importance of specific features, and developing a more robust neural network through targeted tuning and regularization.

We were able to successfully implement these planned changes. In preprocessing, negative and extreme `fire_weather_index` values were clipped to ensure physical validity. Two additional interaction features were engineered:

temp_humidity_interaction: computed as $\text{temp_mean} * (1 - \text{humidity_min} / 100)$.

wind_fwi_interaction: computed as $\text{wind_speed_max} * \text{fire_weather_index}$.

were engineered to better capture the combined effects of meteorological variables on wildfire risk. These features proved especially helpful in distinguishing between medium- and high-risk cases.

Our plan evolved slightly throughout the course of the semester. Initially, we considered experimenting with deeper or hybrid models; however, we ultimately chose to keep the project scope narrow and focus on maximizing the efficiency of a neural network applied to the base dataset. The neural network architecture from earlier iterations was significantly modified, resulting in a more stable

and computationally efficient design, as detailed in the *Implementation* section.

Overall, our final model remains consistent with our original objectives while reflecting a refined and more focused technical approach. The adjustments made to preprocessing and network design substantially improved model stability and interpretability, demonstrating a successful follow-through on our original project plan.

7 Error Analysis

The error analysis for this project included accuracy, per-class precision, recall, and F1-score calculations, as well as a confusion matrix to visualize prediction behavior across risk levels.

7.1 Quantitative Performance

Metric	Value
Accuracy	0.9432
Macro Precision	0.943
Macro Recall	0.933
Macro F1-score	0.939

Table 1: Overall model performance metrics.

Class	Precision	Recall	F1-score
Low Risk (0)	0.96	0.97	0.96
Medium Risk (1)	0.94	0.92	0.93
High Risk (2)	0.93	0.91	0.92

Table 2: Per-class precision, recall, and F1-scores.

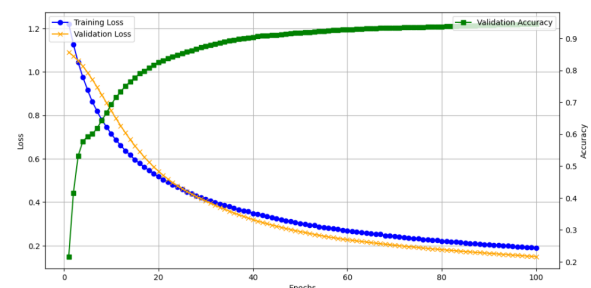


Figure 3: Training and validation loss and validation accuracy across epochs.

True Class	Predicted Class		
	Low	Medium	High
Low	4671	112	18
Medium	97	4274	221
High	21	189	4043

Figure 4: Confusion matrix of wildfire risk classification model.

The confusion matrix (Figure 4) shows that most predictions lie along the diagonal, indicating accurate classification across all classes.

7.2 General Strengths

Our model is very good at identifying low-risk conditions, with a recall of 0.97. While this is not as valuable as high accuracy of high-risk identification, in a real-world use case it would prevent the need for time and resources to be used to further investigate cases that are clearly not a priority. The model is also very adept at handling major meteorological drivers such as humidity and pressure, producing consistent predictions across varied regions. Lastly, the use of regularization and early stopping ensures our model does not excessively overfit.

7.3 Qualitative Error Patterns

When examining specific cases causing errors, we found the following patterns:

- The model somewhat struggles with samples near the decision thresholds, specifically for medium and high-risk levels ($\text{fire_weather_index} \approx 10$ or 30). These samples were the most frequently misclassified, which makes intuitive sense as the two categories share overlapping meteorological characteristics. This is only problematic if medium-risk cases are not treated with proper urgency.
- Additionally, samples from regions with small sample sizes showed slightly higher error rates. This can be explained as a simple sampling error.
- Similarly, samples with meteorological observations that strayed significantly from the mean (such as wind speed or dewpoint) occasionally led to incorrect medium-risk predictions when the true label was low risk. These cases may reflect rapid changes in weather patterns but are not overly problematic, as they are essentially lesser false negatives.

7.4 Future Changes

If we were to continue developing this model, the following improvements could be implemented. The current rigid thresholds at $\text{FWI} = 10$ and 30 could be replaced with quantile- or probability-based thresholds, better reflecting the gradual nature of environmental transitions. Clustering based on geographical region may also help mitigate some of the concerns around locational outliers.

Lastly, using short-term rolling averages for certain features would help stabilize errors caused by erratic but temporary weather conditions.

Team Contributions

- **Eric Solak:** Created Team Contract. Authored and completed Project Proposal sections *Overview*, and *Task Definition*. Set up the GitHub repository, including modularization, project structuring, separation of concerns. Authored and completed Project Progress Report sections 1 *Introduction*, 2 *Related Work*, 3 *Dataset*, 4 *Features*, and 7 *Feedback and Plans*. Implemented final neural network architecture and fixed data leakage to achieve 90% accuracy in final model.
- **Tyler Yue:** Authored and completed Project Proposal section *Results and Evaluation*. Coded parts of the model including the selection of layers in the neural network and adding a class weighting to improve accuracy by modifying the loss penalty for classes with less datapoints. Completed sections 4 and sections 5 of the report and updated to meet new model specifications and results. Tuned parameters of the model to increase accuracy including adjusting the epochs patience and activation functions. Created diagrams for inclusion in report.
- **Ahmed Sahi:** Authored and completed Project Proposal section *Implementation*. Coded the first iteration of the model, including the preprocessing steps in notebook 1, the neural network setup in notebook 2, and the evaluation metrics in notebook 3. Authored and completed Project Report sections *Progress* and *Error Analysis*.

References

- Daniela Cisneros, Jordan Richards, Ashok Dahal, Luigi Lombardo, and Raphaël Huser. 2024. [Deep statistical regression for jointly moderate and extreme australian wildfires](#). *Spatial Statistics*.
- Government of Canada. 2025. [Canadian forest fire danger rating system](#).
- Chenglei Hu, Regina Baltazar Bispo, Håvard Rue, Carlos C. DaCamara, Ben Swallow, and Daniela Castro-Camilo. 2025. [Xgboost meets inla: a two-stage spatio-temporal forecasting of wildfires in portugal](#). *Cornell University*.

Maxwell B Joseph, Matthew W Rossi, Nathan P Mitekiewicz, Adam L Mahood, Megan E Cattau, Lise Ann St Denis, R Chelsea Nagy, Virginia Iglesias, John T Abatzoglou, and Jennifer K Balch. 2019. [Deep statistical regression for jointly moderate and extreme australian wildfires](#). *National Center for Biotechnology Information*.

NASA. 2025a. [Fire information for resource management system](#).

NASA. 2025b. [Wildfires and climate change](#). *NASA Science*.

Open-Meteo. 2025. [Free weather api](#).

Dmitrii Shadrin, Svetlana Illarionova, Fedor Gubanov, Ksenia Evteeva, Maksim Mironenko, Ivan Levchunets, Roman Belousov, and Evgeny Burnaev. 2024. [Wildfire spreading prediction using multimodal data and deep neural network approach](#). *Scientific Reports*.

V. R. Vijayaragul. 2025. [Global wildfire dataset](#).